Wind turbine measurement technique – 46400

Assignment AS03 - Power performance calculation

Deadline 23 March 2025 – <u>group report</u>

Introduction

This assignment demonstrates how the wind turbine power performance is determined based on a reference dataset of power measurements from a 850 kW wind turbine with reference to the IEC 61400-12-1:2017 (ed.2) standard.

Input data

The measurements were recorded on a 850 kW wind turbine with a rotor diameter of 52 m and a hub height of 44 m. The wind turbine's cut-out wind speed is 25 m/s. The measurement data (10-minute statistics) is provided in file *"2023-Jan-July-Power-data_AS03.csv"*. The channel (header) definitions are given in Annex 1.

Power performance calculation

Q3.1: Determine and document the valid measurement sector for the met mast (MM1) and wind turbine under test V52-850 kW located at Risø Campus, according to Annex A and Annex B of ref.1

a) Identify the measurement sector with undisturbed inflow for both met mast (MM1) and wind turbine under test V52.

b) Is a site calibration required? Why?

Q3.2: Determine the filtered and normalized power curve based on data recorded during January - July 2023.

a) Perform data normalization and report the mean air density at the site.
b) Report the bin-averaged values of mean wind speed, mean power, standard deviation of power, Cp-coefficient, number of observations, as well as the category A, s_i, category B, u_i, and combined, u_{ci}, uncertainties for each bin i in tables and plots.

Q3.3: Calculate and report the results of AEP-measured and AEP-extrapolated using Rayleigh wind speed distributions with average annual wind speed at hub height V_{ave}=4, 5, 6, 7 8, 9, 10 and 11 m/s, together with their corresponding (absolute and relative) uncertainties and complete/incomplete labels. The AEP uncertainty should be calculated according to Ref. 1, Annex E, based on parameters listed in Annex 2.

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References

1. IEC 61400-12-1:2017 "Wind turbine power performance testing" (can be downloaded from DTU Library).
2. ENV 13005 guide to the expression of uncertainty in measurement (see JCGM_100_2008_E.pdf)
3. Introduction_AS03_2025.pdf.

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Annex 1*: Available 10-minute statistics in 2023-Jan-July-Power-data_AS03.csv file*

Name	Unit	Height[m]	Description
date_time	-	-	Runname = timestamp with reference to the starting time for the 10 minute period.
ActPow	kW	-	Mean active power
ActPow_stdev	kW	-	Stdev of active power
ActPow_min	kW	-	Minimum active power
ActPow_max	kW	-	Maximum active power
ROT	rpm	-	Mean rotor speed
ROT_stdev	rpm	-	Stdev of rotor speed
ROT_min	rpm	-	Minimum rotor speed
ROT_max	rpm	-	Maximum rotor speed
Pitch	deg	-	Mean pitch angle
Pitch_stdev	deg	-	Stdev of pitch angle
Pitch_min	deg	-	Minimum pitch angle
Pitch_max	deg	-	Maximum pitch angle
yaw	deg	-	Mean yaw angle
yaw_stdev	deg	-	Stdev of yaw angle
Wsp_44m	m/s	44	Mean wind speed. Mounted on South boom
Wsp_44m_stdev	m/s	44	Stdev of wind speed. Mounted on South boom
TI_44m	%	44	Mean turbulence intensity. Mounted on South boom
Wdir_41m	deg	41	Mean wind direction. Mounted on North boom
Wdir_41m_stdev	deg	41	Stdev wind direction. Mounted on North boom
AirAbs_70m	degC	70	Mean temperature. Mounted on South boom
Press_enc_2m	hPa	2	Mean atm. pressure, measured in mast
RH_2m	%	2	Mean relative humidity

Approximations:
- use the wind direction at 41m as a proxy for the wind direction at hub height
- use the pressure and humidity at 2m as proxies for the pressure and humidity at hub height.

***Annex 2**: Parameters used to estimate the combined uncertainty of AEP*

Item	Uncertainty, u	Sensitivity factor
Category B uncertainties in Electric power	$u_{P,i} = \sqrt{((0.002 \times P_i)^2 + 3.7^2 + 0.3^2)}$	1
Category B uncertainties in wind speed	$u_{V,i} = \sqrt{(0.025^2 + (0.038+0.0038xV_i)^2 + (0.01xV_i)^2 + (0.02xV_i)^2 + (0.001*V_i)^2)}$	$c_{V,i} = \lvert P_i - P_{i-1}\rvert / \lvert V_i - V_{i-1}\rvert$
Category B uncertainties in air temperature	$u_{T,i}$= 0.6°K	$c_{T,i} = c_{V,i} * V_i/(3*288.15)$
Category B uncertainties air pressure	$u_{B,i}$=2.0hPa	$c_{B,i} = c_{V,i} * V_i/(3*1013)$
Category B uncertainties relative humidity	$u_{RH,i}$= 0.63%RH	$c_{RH,i} = c_{V,i} * V_i * 0.0018$

Annex 3: *Test site and coordinates*

	WGS84, UTM33		Dimensions	
	E [m]	N [m]	Hub height [m]	Rotor diamenter [m]
Mast (MM1)	317,438	6,175,028		
Turbine (V52)	317,548	6,174,985	44	52
V27	317,519	6,174,901	31.5	27

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Reporting

Questions	Weight	Name#1	Name#2	Name#3
Q4.1	30%			
Q4.2	30%			
Q4.3	40%			

Please enter names and mark your contributions as percentage of total work performed for each question.

All groups larger than one person must attach to the report as an additional page a small progress report signed by all members of how the group worked together, and the work load for each member MUST explicitly be stated as a percentage. You cannot pass the course unless this is done.

Required: *An assignment report of 15 – 20 pages including a table with student contributions. The report should present the problem(s), recordings from lab or from site visits and how the problems have been solved. Furthermore, each question should be addressed clearly and discussed when this is necessary.*

Q 4.1: Weight 30%

Q 4.2: Weight 30%

Q 4.3: Weight 40%